

```
# import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# load the dataset
df = pd.read_csv('hotel_bookings.csv')
```

```
# Preview our dataset
df.head()
```

[Show hidden output](#)

```
# Show the shape of the dataset (rows and columns)
df.shape
```

(119390, 32)

```
# Overview of the columns and data types
df.info()
```

[Show hidden output](#)

```
# check for missing values
df.isnull().sum()
```

[Show hidden output](#)

## ▼ Data Cleaning

```
# Handle missing values
# 0 indicates booked directly no agency / company
df['company'] = df['company'].fillna(0)
df['agent'] = df['agent'].fillna(0)
```

```
# Drop missing values in country
df = df.dropna(subset=['country'])
```

```
# Fill missing 'children' with 0
df['children'] = df['children'].fillna(0)
```

```
# Fix incorrect data types
# Converts floats back to integers
df['children'] = df['children'].astype(int)
df['agent'] = df['agent'].astype(int)
```

```
# Remove duplicates
df = df.drop_duplicates()
```

```
# Verify no missing values remain
df.isnull().sum()
```

[Show hidden output](#)

## Feature Engineering

```
# Total nights
df['total_nights'] = df['stays_in_week_nights'] + df['stays_in_w
```

```
# Total Guests
df['total_guests'] = df['adults'] + df['children'] + df['babies
```

```
# is family - (True if there are children or babies, False other
df['is_family'] = (df['children'] > 0) | (df['babies'] > 0)
```

```
# Season
# Mapping months to seasons
month_to_season = {
    'December': 'Winter', 'January': 'Winter', 'February': 'Win
    'March': 'Spring', 'April': 'Spring', 'May': 'Spring',
    'June': 'Summer', 'July': 'Summer', 'August': 'Summer',
    'September': 'Fall', 'October': 'Fall', 'November': 'Fall'
}
```

```
df['season'] = df['arrival_date_month'].map(month_to_season)
```

```
# Drop rows where total_guests is 0 (an invalid booking)
invalid_bookings = df[df['total_guests'] == 0].shape[0]
invalid_bookings
```

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```
# drop invalid_bookings
df = df[df['total_guests'] > 0]
```

```
# to csv file the cleaned data
df.to_csv('hotel_bookings_cleaned.csv', index=False)
```

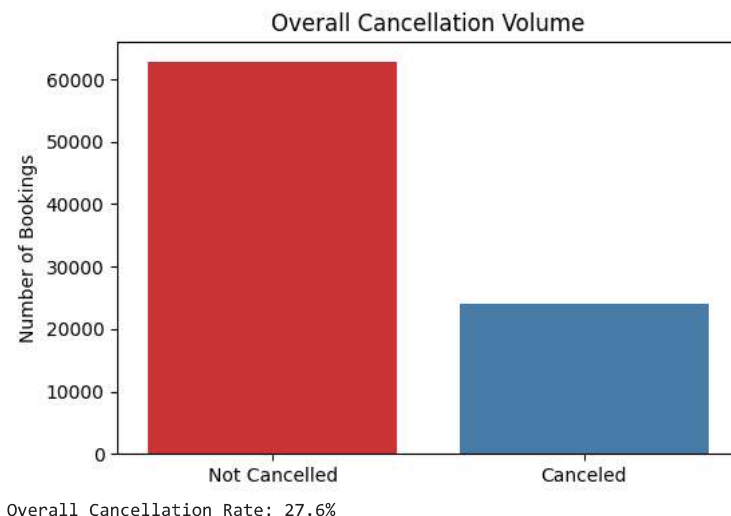
## EDA

```
# Overall cancellation distribution
plt.figure(figsize=(6, 4))
# mapping 0 and 1 to readable labels for the chart
df_plot = df.copy()
df_plot['is_canceled_label'] = df_plot['is_canceled'].map({0: 'Not Canceled', 1: 'Canceled'})

sns.countplot(x='is_canceled_label', data=df_plot, hue='is_canceled_label')
plt.title('Overall Cancellation Volume')
plt.ylabel('Number of Bookings')
plt.xlabel('')
plt.savefig('overallcancellation.png')
plt.show()

# Calculate exact percentage
cancel_pct = df['is_canceled'].value_counts(normalize=True) * 100

print(f"Overall Cancellation Rate: {cancel_pct[1]:.1f}%")
```

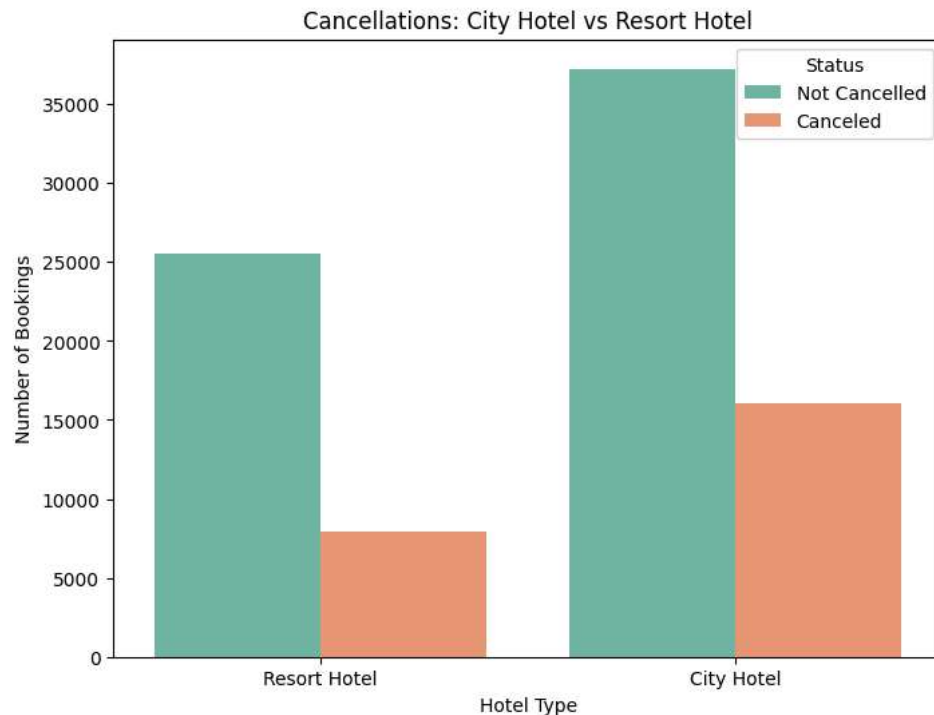


The overall cancellation rate is 27%- This means that roughly 1 in 3 bookings ends up being cancelled. This is a massive loss pipeline for the hotels

```
# Hotel Type Comparison
# Does the type of hotel affect the likelihood of cancellation?
plt.figure(figsize=(8,6))
sns.countplot(x='hotel', hue='is_canceled_label', data=df_plot,
plt.title('Cancellations: City Hotel vs Resort Hotel')
plt.ylabel('Number of Bookings')
plt.xlabel('Hotel Type')
plt.legend(title='Status')
plt.savefig('hoteltype.png')
plt.show()

# Calculation rates per hotel type
```

```
print(df.groupby('hotel')['is_canceled'].value_counts(normalize=
```



```
hotel    is_canceled
City Hotel 0      69.901072
           1      30.098928
Resort Hotel 0     76.289090
            1     23.710910
Name: proportion, dtype: float64
```

City Hotels see significantly higher booking volumes but they also suffer from a much higher cancellation rate compared to Resort Hotels.

City trips are often short, impulsive, or tied to business meetings that get moved. Resort trips are usually planned vacations (often involving flights) making guests much less likely to abandon their plans.

```
# Monthly Trends(Seasonality)
```

```
plt.figure(figsize=(12,6))
```

```
# define chronological order for months
```

```
month_order = ['January', 'February', 'March', 'April', 'May',
               'August', 'September', 'October', 'November', 'December']
sns.countplot(x='arrival_date_month', hue='is_canceled_label',
              data=df_plot, order=month_order, palette='muted')
```

```
plt.title('Booking Volume and Cancellation by Month')
```

```
plt.ylabel('Number of Bookings')
```

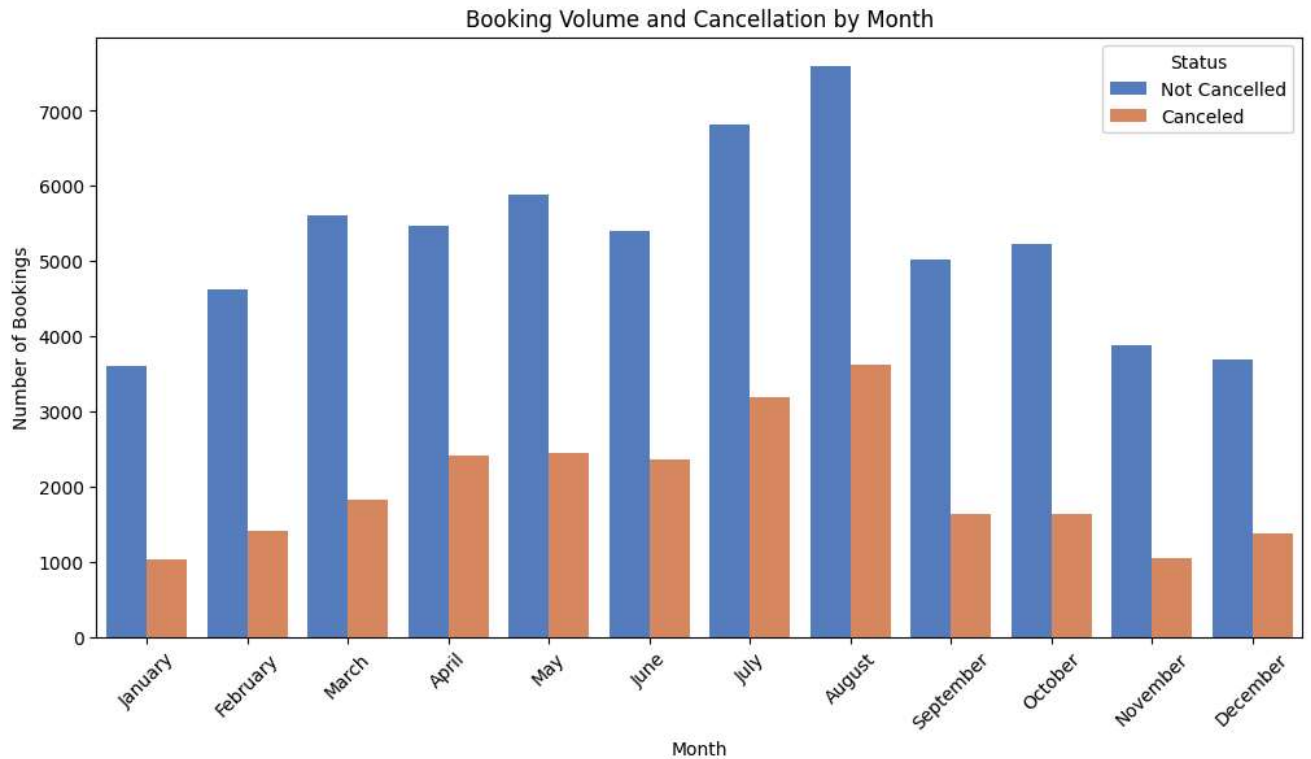
```
plt.xlabel('Month')
```

```
plt.xticks(rotation=45)
```

```
plt.legend(title='Status')
```

```
plt.savefig('monthlytrends.png')
```

```
plt.show()
```



July and August are our peak months, aligning with the European summer holiday season. Because demand naturally outstrips supply during these months the hotel has leverage.

Recommendation:

The hotel should implement stricter, non-refundable booking rates during the summer because they can easily find replacement guests.

### # Lead Time Analysis

# Lead time is the number of days between when the a guest book:

# Does booking in advance make a guest more likely to cancel

```
plt.figure(figsize=(8,5))
```

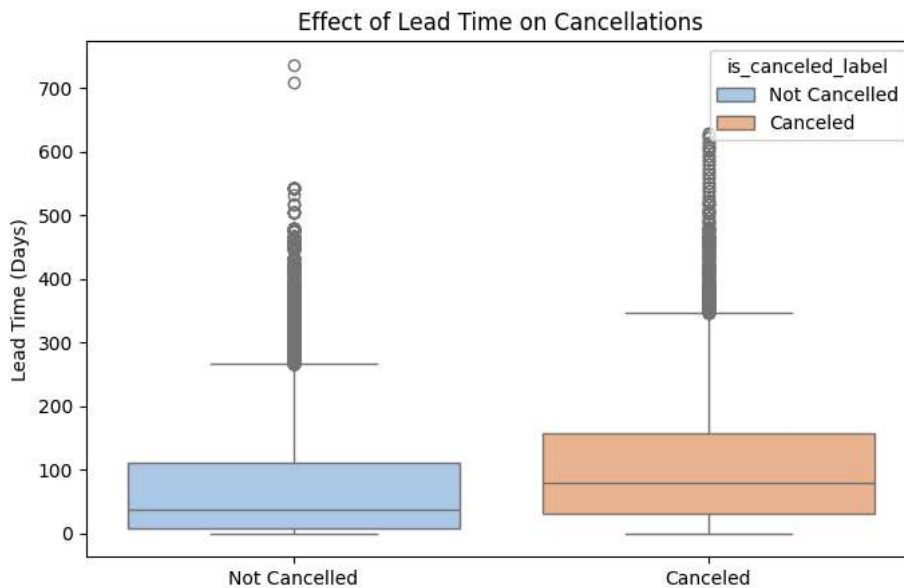
```
sns.boxplot(x='is_canceled_label', y='lead_time', data=df_plot,
plt.title('Effect of Lead Time on Cancellations')
```

```
plt.ylabel('Lead Time (Days)')
```

```
plt.xlabel('')
```

```
plt.savefig('leadtime.png')
```

```
plt.show()
```



The boxplot clearly shows that the lead time for canceled booking is much higher. – Basically: the earlier someone books, the more time they have to change their mind, find a better deal or experience a schedule change.

Recommendation:

Introduce a progressive deposit system where bookings made more than 90 days in advance require a 20% non refundable deposit

```
# Country Distribution
# Top 10 Countries
plt.figure(figsize=(10,6))
# Get top 10 countries by volume
top_10_countries = df['country'].value_counts().head(10).index

sns.countplot(x='country', hue='is_canceled_label', data = df_p,
              ,order = top_10_countries, palette='Dark2')
plt.title('Top 10 Countries by Booking Volume')
plt.ylabel('Number of Bookings')
plt.xlabel('Country Code')
plt.legend(title='Status')
plt.savefig('countrydistribution.png')
plt.show()
```

Top 10 Countries by Booking Volume



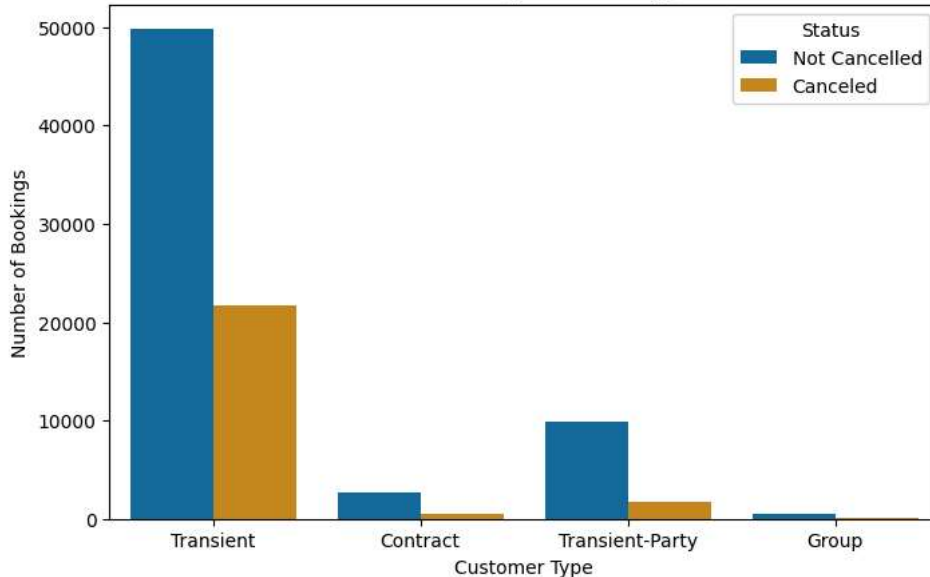
PRT(Portugal) generate the highest volume of bookings, but it also has a staggering cancellation rate of 50%. Because the hotels in this dataset are located in Portugal, these are local residents. Locals often book multiple hotels as "backups" for weekend getaways because they don't have to commit to flights. International guests(GBR, FRA, DEU) rarely cancel because their travel plans are locked in by airline tickets.

### # Customer Type Analysis

# Are solo/transient travellers more volatile than large groups

```
plt.figure(figsize=(8,5))
sns.countplot(x='customer_type', hue='is_canceled_label', data=)
plt.title('Cancellation by Customer Type')
plt.ylabel('Number of Bookings')
plt.xlabel('Customer Type')
plt.legend(title='Status')
plt.savefig('customertype.png')
plt.show()
```

Cancellation by Customer Type



Transient Travellers (Independent/Solo traveller not part of a group) account for the vast majority of bookings and cancellations. Conversely, 'Groups' and 'Contract' bookings are much more stable.

Recommendation: The sales team should prioritize securing more B2B contracts and group events to build a stable, guaranteed revenue base that protects against the volatility of Individual Transients.